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Non-Attacking Black and White Knights

Difficulty: **Medium**Accuracy: **51.2%**Submissions: **35K+**Points: **4**

Given two integers **n** and **m** representing the dimensions of a chessboard, find the number of ways to place one black knight and one white knight on the chessboard such that they cannot attack each other.



Note:

- The knights have to be placed on different squares.
- A knight can move two squares horizontally and one square vertically (L shaped), or two squares vertically and one square horizontally (L shaped).
- The knights attack each other if one can reach the other in one move.

Examples:

Input: $n = 2, m = 2$ **Output:** 12**Explanation:** There are 12 ways we can place a black and a white Knight on this chessboard such that they cannot attack each other.**Input:** $n = 2, m = 3$ **Output:** 26**Explanation:** There are 26 ways we can place a black and a white Knight on this chessboard such that they cannot attack each other.

Constraints:

 $1 \leq n \leq 200$ $1 \leq m \leq 225$ [Try more examples](#)

Expected Complexities



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```
1 class Solution:
2     def numOfWays(self, n, m):
3         total = n * m * (n * m - 1)
4
5         attack_pairs = 0
6
7         if n > 1 and m > 2:
8             attack_pairs += 2 * (n - 1) * (m - 2)
9
10        if n > 2 and m > 1:
11            attack_pairs += 2 * (n - 2) * (m - 1)
12
13        return total - 2 * attack_pairs
```

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